# STATEMENT OF PURPOSE

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From a young age, I have been deeply committed to finding solutions to have an equitable and peaceful society, one where freedom is cherished, and individuals of all backgrounds can thrive. I have always strived to understand how we can improve our world for the current and future generations, ensuring that they can live in a society where the value of privacy, and equality is not mere ideals, but an achievable reality. This idea was reinforced by the time I became familiar with cryptography and its applications to bring privacy and security across digital systems. Particularly, the revolutionary appearance of blockchain technology has inspired me, a solution to bring more control from centralized authorities to users, while bringing more trust to the system. Moreover, blockchain acted as a pivotal bridge, sparking my interest in distributed systems and distributed algorithms. I believe that utilizing computer science concepts to design protocols that enhance the security and privacy of systems has the potential to yield groundbreaking advancements, a path I aspire to pursue as a Ph.D. student at the University of Southern California, collaborating with top experts in this domain.

**Research in Blockchain Technology and Distributed Systems.** To pursue my initial interest in blockchain and cryptocurrencies, in the summer of 2022, I joined the Data Analytic lab at the University of Tehran as a research assistant working with Prof. Behnam Bahrak on designing a new blockchain consensus protocol. First, I developed expertise in foundations of cryptography, distributed systems, and blockchain technology, building up knowledge to discover current limitations and potential research directions. One of the arch problems with Bitcoin and many other cryptocurrencies is their power consumption. Bitcoin is overkill because its consensus protocol has a very expensive building block, and its transaction throughput is much lower than centralized payment networks. After reviewing the state-of-the-art literature, we found PoA (Proof of Activity), published in 2014, to be a promising idea for further enhancements. We started to work on applying the user's activities on the blockchain to their mining chance. Then, we defined the activity parameter based on the contributions of users to the system and added it to the mining formula of the blockchain. Therefore, users who are more active in the system have a greater chance of mining the next block, indicating that mining does not solely depend on the miners' hash power. My main contribution was designing the activity system mathematically, in a way to ensure the resilience and security of the system against malicious attackers, besides maximizing honest users' profits. Also, I proposed an evaluation based on game-theory concepts for our protocol, similar to the Prisoner's Dilemma problem. Our solution incentivizes users to increase their honest contributions to maximize their profit. Consequently, the Nash equilibrium converges to the point that not only enhances the target system's security and stability but also makes it more dynamic and full of worthwhile activities. This scheme can be applied to various cryptocurrency networks, and to practically assess our scheme, we implemented a distributed file-sharing system powered by this protocol as a proof of concept, called Crystalline. The output manuscript of this work is currently under review by the IET Blockchain journal.

**Research in Security and Privacy.** Having experience in cryptography and blockchain technology, I decided to do a research internship at the ACIS lab at the University of Victoria, under the supervision of Prof. Homayoun Najjaran to work on privacy preservation techniques in cryptocurrencies. My focus was on exploring mixing and obfuscation techniques within cryptocurrencies, a domain predominantly reliant on cryptography and graph theory methods to enhance users' privacy and anonymity. While numerous mixing services have been proposed in both academia and industry, we discerned a notable absence of an in-depth analysis and evaluation of these services. Therefore, we initiated an exploration of mixing solutions, with the aim of writing a comprehensive survey paper on this topic, in which I was the principal author. To illustrate our research methodology, I first delved into the existing literature on mixing services, allowing us to shape the paper's structure. Next, the research procedure followed by identifying and defining the essential mixing functionalities crucial for ensuring anonymity. Subsequently, I conducted a thorough analysis of over 32 different mixing services, drawing insights from both academic and real-world solutions. Through a comparative analysis of these two realms, I underscored the distance between academia and industry, illuminating the possibilities for mutual enhancement on both sides through their integration. Ultimately, our methodology culminated in the development of an evaluation framework designed to assess the effectiveness of mixing services, while also highlighting the open problems and research challenges in this domain. These experiences equipped me with skills to tackle complex issues systematically, a valuable asset for future research projects as a Ph.D. student at the University of Southern California. My first-author manuscript resulting from this research is currently in the review of the Journal of Network and Computer Applications.

In an era where digital systems are increasingly integrated into our daily lives, the storage and sharing of personal information online have become commonplace. As we entrust our data to these systems, the imperative to safeguard it from unauthorized access and misuse has become paramount. Moreover, with the rapid pace of revolutionary innovations driven by artificial intelligence (AI), the realm of security and privacy emerges as a central concern for the future, and I believe that cryptography could be the solution to address these concerns, presenting a valuable research opportunity in my perspective. Also, I have a strong interest in conducting research on distributed systems and algorithms due to their ability to offer scalable, consistent, fault-tolerant solutions for modern computing challenges. Therefore, with my vision truly in mind, I dedicate the prime years of my life to conducting research deeply to become a master in my field, aiming to discover unknown research territories and contribute meaningfully to society.

USC Viterbi School of Engineering is my top destination for my Ph.D. studies due to its strong and prestigious research groups in security, privacy, and distributed system domains. In particular, my research interests align well with Prof. Sirvastan Ravi's research on distributed algorithms, and his project on concurrency and security of smart contracts is very fascinating. In addition, Prof. Seo Jin Park's research on distributed systems and also flash burst computing is very interesting. Also, I am interested in working with Prof. Cyrus Shahabi on topics related to fairness and differential privacy. In the long term, I want to continue my path as a researcher, and the level of expertise I can earn from USC Viterbi and its bright professors/students will help me to reach my goal of working in top-notch research institutes and industries.